

# Abeer Sethia

+91-9148019367 | [abeersethia3105@gmail.com](mailto:abeersethia3105@gmail.com) | [linkedin.com/in/abeer-sethia](https://linkedin.com/in/abeer-sethia) | [abeersethia.github.io](https://abeersethia.github.io)

Computer Vision and Deep Learning Engineer specializing in multimodal perception, 2D–3D reconstruction, and physics-informed deep learning. Developed **structure-recovering signal models** at IIT Kharagpur, engineered **real-time production inference pipelines** at Philips, and **advanced self-supervised trajectory reconstruction** at IISc Bangalore.

## EDUCATION

Manipal Institute of Technology, Manipal

2022–2026

*Bachelors of Technology - Data Science and Engineering, Minor in Multimodal Intelligent Systems*

**Courses:** Computer Vision, Natural Language Processing, Deep Learning, Mathematical Foundations, Statistics

## SKILLS

- **Deep Learning:** CNNs, RNNs, LSTMs, GNNs, Attention Mechanisms & Transformers, U-Nets, VLMs
- **Computer Vision:** Object Detection, Segmentation, Camera Geometry, Optical Flow, Classification
- **Natural Language Processing:** Named Entity Recognition (NER), Multi-Label Sentiment Analysis, NLI Ensembling
- **Frameworks & Tools:** PyTorch, PyTorch3D, PyG, TensorFlow, Keras, CUDA, OpenCV, ITK, MONAI, Git, Docker
- **Languages:** Python, MATLAB, C++, Java, R, Bash (Linux), SQL, L<sup>A</sup>T<sub>E</sub>X
- **Distributed Systems:** Hadoop, Apache Spark, Hive, Pig

## PROFESSIONAL EXPERIENCE

Technical Intern, Philips

March–July 2025

- Engineered a real-time **Flask inference pipeline** for liver lesion segmentation, deploying fine-tuned **nnU-Net** and **MedSAM2** models validated on **500+ clinical CT/MRI scans**.
- Achieved **0.92 Dice (0.89 IoU)**, outperforming standard U-Net baselines by **12%**.

## RESEARCH EXPERIENCE

Research Intern – AI, UTSAAH Lab, Indian Institute of Science Bangalore

December 2025–Present

- Developing a trackerless fetal ultrasound pipeline using **contrastive self-supervised learning** to extract robust temporal representations from B-mode sequences.
- Engineering an  **$SE(3)$  pose estimation network** to predict pose from short frame windows, validating trajectories against **optical and EMT tracker ground truth**.

Research Intern - AI, KLIV Lab, Indian Institute of Technology Kharagpur

April–October 2025

- Engineered a temporal CNN autoencoder to **reconstruct 3D chaotic attractor dynamics from a single-channel** observable, achieving **98.3% (X/Y)** and **93.4% (Z)** phase-space correlation at an 8:1 compression ratio.
- Formulated a **differentiable False Nearest Neighbors (FNN) loss** function to enforce causal consistency and global invariants, preventing trajectory self-intersection during multi-dimensional embedding trajectory recovery.

## PROJECTS

DropEdge++ with Johnson–Lindenstrauss Projection (Minor Specialization Project)

- Developed a GNN framework combining **topology-aware edge dropout** (DropEdge++) with mathematical-based random feature projection to mitigate oversmoothing and improve computational efficiency.
- Achieved up to **8% accuracy gains** and **3× faster training** on the **Cora citation network benchmark**.

HVQ-AP: Hippocampal Volume Quantification for Alzheimer’s Progression

- Quantified hippocampal volumetric progression in Alzheimer’s patients from 3D MRI by building an end-to-end pipeline covering DICOM ingestion (PyDICOM), 3D CNN segmentation, and longitudinal volume tracking.
- Achieved **Dice 91.6% and Jaccard 84.6%** on the **Medical Segmentation Decathlon benchmark**, matching clinical-grade validation standards.

BERT-based SandwichFormer for CoLA and IMDB

- Reproduced and extended the SandwichFormer Transformer architecture in PyTorch for linguistic acceptability and sentiment classification, achieving **0.57 MCC (+5 points over BERT baseline)** and **0.86 F1 on CoLA**.
- Built an NLP training and evaluation pipeline for CoLA and IMDB benchmarks, **attaining 88% accuracy** and **0.89 F1 on long-form sentiment analysis** while enabling reproducible experimentation with Transformer variants.

## CO/EXTRA CURRICULAR

- **Expertise Head – Research Society Manipal:** Mentored 20+ students in research and scientific writing.
- **Data Analyst – IAESTE LC Manipal:** Built automation workflows and dashboards for **1,000+ members**, reducing manual effort by **70%** and improving outreach efficiency by **30%**.